

LAB WORKSHEET

Lab 2 — Storage Deployment: RAID 0/1/5/6/10 with mdadm

Name		Date	
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Objective: 1.2 — Deploy and manage storage: RAID 0/1/5/6/10, JBOD, hardware vs. software, capacity planning.

Work through the lab on the Killercoda Ubuntu playground and record your results below.

Task 1 — Build the loopback disks

Create six 200M loopback files and attach them.

Paste `losetup -a` output showing `/dev/loop1-6`.

Task 2 — RAID 5 capacity

Create the RAID 5 array and read its Array Size.

Record the usable capacity and confirm it equals $(N-1) \times \text{disk}$.

Task 3 — RAID 10 capacity

Create the RAID 10 array.

Record usable capacity and confirm it equals $N/2 \times \text{disk}$.

Task 4 — Fail and rebuild

Fail one disk in a mirror, add a replacement and watch `mdstat`.

Paste the `/proc/mdstat` `resync` line and note the rebuild time.

Task 5 — Level selection

Decide the best RAID level for a large-SATA archive vs. a database.

State your choice for each and one reason.

Completion checklist

- Built RAID 0, 1, 5, 6, 10 and JBOD on loopback disks.
- Recorded the correct usable-capacity figure for each level.
- Failed a disk and observed a successful rebuild.
- Stated when RAID 6 beats RAID 5 and when RAID 10 beats both.
- Compared hardware vs. software RAID trade-offs.